

# Question Bank



**MALLA REDDY  
UNIVERSITY**

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## **Cloud Computing**

**III B.Tech I Semester**

**(2025-26)**

**Department of CSE (Data Science)**

**School of Engineering**



**School of Engineering-A.Y 2025-2026**

**Regulation: R23**

**(CSE-DATA SCIENCE)**

**III Year B. Tech - I Semester**

**Subject Name: CLOUD COMPUTING**

**Subject Code: MR23-1CS0307**

**QUESTION BANK**

| Qno | Question                                                                                                                                                            | Marks  | Section    |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------------|
| 1   | Define Cloud Computing? Compare the differences between Traditional Computing and Cloud Computing?                                                                  | 8      | Section-I  |
| 2   | Summarize the four roots of Cloud Computing in your own words.                                                                                                      | 8      | Section-I  |
| 3   | Demonstrate how two Internet technologies can be used to implement cloud-based services.                                                                            | 8      | Section-I  |
| 4   | Elaborate Cloud service models with example.                                                                                                                        | 8      | Section-I  |
| 5   | a) Analyse Cloud Computing Challenges?<br>b) What are the features of Cloud Computing?                                                                              | 4<br>4 | Section-I  |
| 6   | Discuss the characteristics and purposes of each cloud deployment model.                                                                                            | 8      | Section-I  |
| 7   | a) Explain about Infrastructure of a service provider<br>b) Discuss the key features and functions of Platform as a Service (PaaS) providers.                       | 4<br>4 | Section-I  |
| 8   | Illustrate the different types cloud with an Example?                                                                                                               | 8      | Section-I  |
| 9   | a) What are the Cloud Challenges and risks of Cloud Computing, Explain in detail.<br>b) Illustrate the difference between Traditional Computing vs Cloud Computing? | 4<br>4 | Section-I  |
| 10  | Explain about seven step model for cloud migration in detail with neat diagram?                                                                                     | 8      | Section-I  |
| 11  | What is AWS? Explain about AWS Cloud Adaptation Framework in detail?                                                                                                | 8      | Section-II |
| 12  | a) Discuss the main service categories offered by AWS and their functions.<br>b) Explain Service categories of AWS                                                  | 4<br>4 | Section-II |
| 13  | State the main components of AWS Global Infrastructure and explain how they work together.                                                                          | 8      | Section-II |
| 14  | a) Explain the service categories of AWS in detail?<br>b) Summarize the importance of the AWS Cloud Adoption Framework in guiding cloud adoption.                   | 4<br>4 | Section-II |
| 15  | Briefly Explain about Importance of Services and                                                                                                                    | 8      | Section-II |

|    |                                                                                                                                                                                              |        |                    |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------------------|
|    | Service categories.                                                                                                                                                                          |        |                    |
| 16 | List the design principles of the AWS Well-Architected Framework.                                                                                                                            | 8      | <b>Section-II</b>  |
| 17 | a) Describe the Operational Excellence pillar and explain its design principles in AWS.<br>b) Explain about Performance Efficiency pillar with design principles in AWS                      | 4<br>4 | <b>Section-II</b>  |
| 18 | Discuss the difference between reliability and high availability with examples.                                                                                                              | 8      | <b>Section-II</b>  |
| 19 | What are the main design principles of four AWS pillars?                                                                                                                                     | 8      | <b>Section-II</b>  |
| 20 | a) What are the main security principles in AWS?<br>b) Describe how the Security pillar helps protect cloud workloads and data.                                                              | 4<br>4 | <b>Section-II</b>  |
| 21 | Illustrate the AWS Security Shared Responsibility Model with a diagram and explain the responsibilities of AWS and the customer.                                                             | 8      | <b>Section-III</b> |
| 22 | What is AWS IAM? What are the components essential components in IAM and what is the purpose of IAM in AWS                                                                                   | 8      | <b>Section-III</b> |
| 23 | a) State the importance of AWS IAM in cloud security.<br>b) How to provide the security to multiple accounts and data.                                                                       | 4<br>4 | <b>Section-III</b> |
| 24 | List the main purposes of AWS CloudWatch.                                                                                                                                                    | 8      | <b>Section-III</b> |
| 25 | Relate the concept of Elastic Load Balancing to its role in distributing traffic across multiple AWS instances.                                                                              | 8      | <b>Section-III</b> |
| 26 | a) What is VPC? How VPC is implemented in Cloud?<br>b) Extend your explanation of VPC security by describing how it can be enhanced with network ACLs, security groups, and VPN connections. | 4<br>4 | <b>Section-III</b> |
| 27 | State the definition of a Virtual Private Cloud (VPC) and identify its key parts in a diagram.                                                                                               | 8      | <b>Section-III</b> |
| 28 | a) Define an Internet Gateway in VPC networking and list its main functions.<br>b) Describe the working of a NAT Gateway and relate it to secure internet access for private resources.      | 4<br>4 | <b>Section-III</b> |
| 29 | Explain about VPC peering and VPC Direct connection with a diagram                                                                                                                           | 8      | <b>Section-III</b> |
| 30 | a) What is Cloud front explain in detail about it?<br>b) Explain about Security groups and Network                                                                                           | 4      | <b>Section-III</b> |

|    |                                                                                                                         |        |                   |
|----|-------------------------------------------------------------------------------------------------------------------------|--------|-------------------|
|    | ACLs                                                                                                                    | 4      |                   |
| 31 | Elaborate AWS compute services.                                                                                         | 8      | <b>Section-IV</b> |
| 32 | Illustrate the different types of AWS EC2 instances and describe their use cases.                                       | 8      | <b>Section-IV</b> |
| 33 | What is AMI? What is the purpose of AMI while creating Instance?                                                        | 8      | <b>Section-IV</b> |
| 34 | a) Illustrate about EC2 vs Virtual Machine?<br>b) Explain about volumes and Snapshots in AWS.                           | 4<br>4 | <b>Section-IV</b> |
| 35 | Demonstrate about ec2 Cost Optimization?                                                                                | 8      | <b>Section-IV</b> |
| 36 | a) Explain the importance of Amazon EC2.<br>b) How to launch a Amazon Instance in AWS.                                  | 4<br>4 | <b>Section-IV</b> |
| 37 | How to create a Lambda function for displaying Hello World                                                              | 8      | <b>Section-IV</b> |
| 38 | Discuss about Serverless computing platform with key characteristics.                                                   | 8      | <b>Section-IV</b> |
| 39 | How does Elastic Beanstalk works in AWS?                                                                                | 8      | <b>Section-IV</b> |
| 40 | Explain about AWS Elastic Beanstalk deployment.                                                                         | 8      | <b>Section-IV</b> |
| 41 | Elaborate about Amazon Block Store                                                                                      | 8      | <b>Section-V</b>  |
| 42 | a) What are the features of EBS?<br>b) State the different types of AWS EBS volumes and their primary use cases.        | 4<br>4 | <b>Section-V</b>  |
| 43 | Discuss about Amazon Simple Storage Service                                                                             | 8      | <b>Section-V</b>  |
| 44 | Demonstrate the different Amazon S3 storage classes and explain when each should be used.                               | 8      | <b>Section-V</b>  |
| 45 | a) Discuss the importance of Amazon EBS.<br>b) Define S3 Life Cycle policies                                            | 4<br>4 | <b>Section-V</b>  |
| 46 | Explain about Amazon RDS instance and How to Create an RDS Instance in AWS?                                             | 8      | <b>Section-V</b>  |
| 47 | a) Explain about AWS RDS Database Engine<br>b) Differentiate the Relational and non-relational data bases               | 4<br>4 | <b>Section-V</b>  |
| 48 | What is Amazon Dynamo DB and its core components?                                                                       | 8      | <b>Section-V</b>  |
| 49 | How does Amazon Redshift simplify data warehouse and analytics management?                                              | 8      | <b>Section-V</b>  |
| 50 | a) State the difference between SQL and NoSQL databases in AWS.<br>b) List the main features of SQL and NoSQL databases | 4<br>4 | <b>Section-V</b>  |